Gain-vs-charge correction for cluster energies

What the problem is

The gain (ADU per electron) is normally measured from the spacing of the 0e- and 1e- peaks and then used for every pixel. But the amplifier gain falls slowly as the charge in a pixel grows: typically ~3% lower at 1000 e- (0.2-8% across 111 Image-5 CCDs). X-ray clusters have pixels with hundreds to thousands of electrons, so their energies come out a few percent too low. On top of that, the 0e-/1e- gain fit of each image can be off by 1-4%, so different CCDs disagree.

What the correction does

1. Measure g(n), the gain at n electrons per pixel, from the images themselves: a skipper image's pixel values form a comb of peaks, one per electron count; the local spacing of that comb at 50, 300, 800 e- ... is the gain at that charge (measured in clean rows, all images of the same module/CCD together). Fit: g(n) = a + b*n, extrapolated up to 3x the highest measured charge, constant beyond.

2. Convert every cluster pixel back to ADU, then to electrons with that curve (instead of one gain), and re-sum the cluster energy. Written as a second set of cluster files (root_gaincorr/); the originals are kept.

3. CCDs whose curve cannot be trusted (comb visible over < 4 charge bins or < 400 e-, or a rising gain) are left unchanged. No source peak is used anywhere.

How to use it

run_WADERs images.txt --gain-correction (or the standalone tool: gain_vs_charge.py measure / apply)

Result over the Production_Modules_Data (tool frozen, nothing tuned on these data)

Image 5, Fe-55 K-alpha of the 82 corrected CCDs in 10 datasets: average 0.9835 -> 0.9977 of the true energy; spread between CCDs 2.1% -> 0.32%.

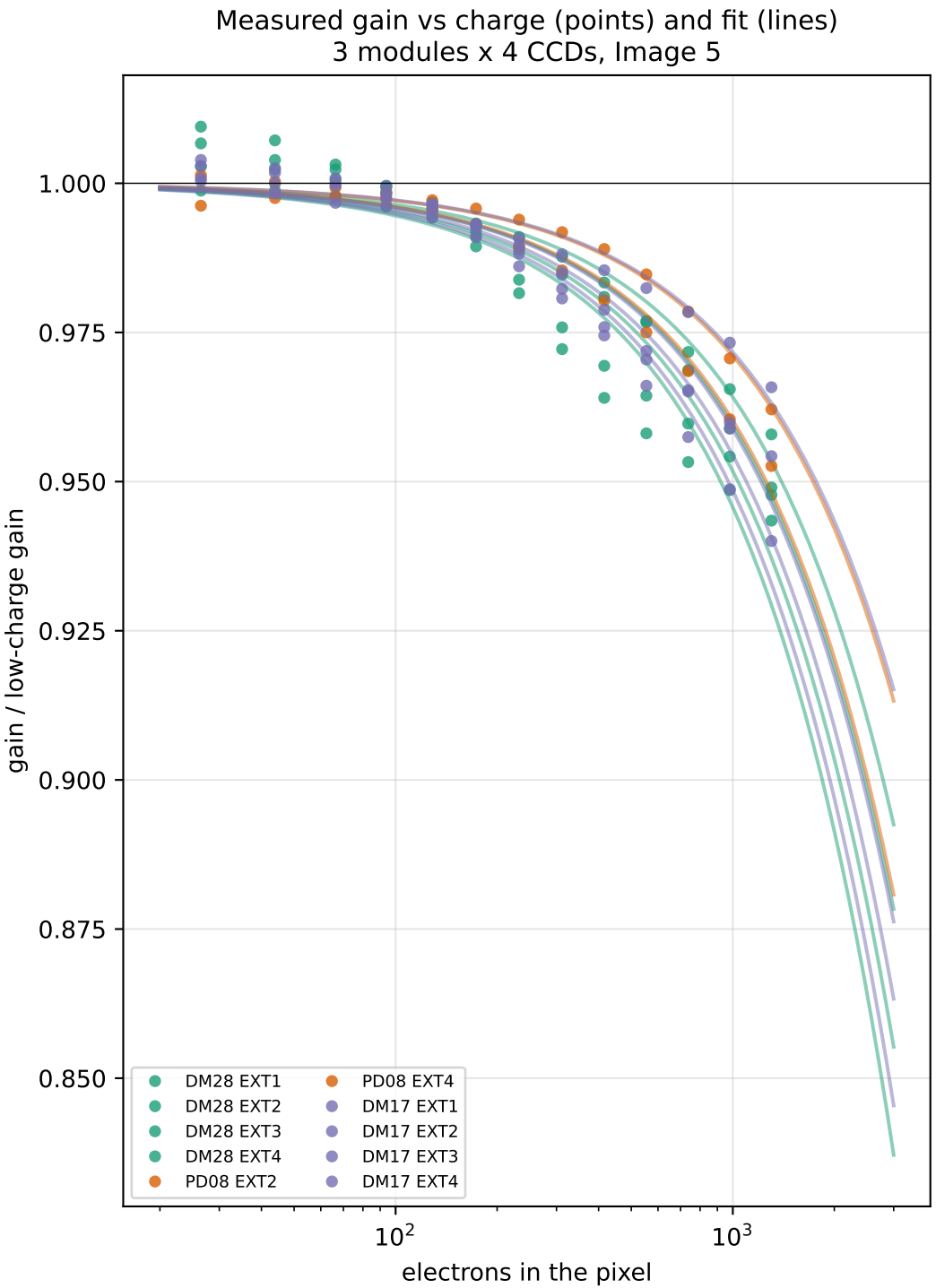
Binned Images 6/7: average ~0.98-0.99 -> 0.998-1.002, spread roughly halved.

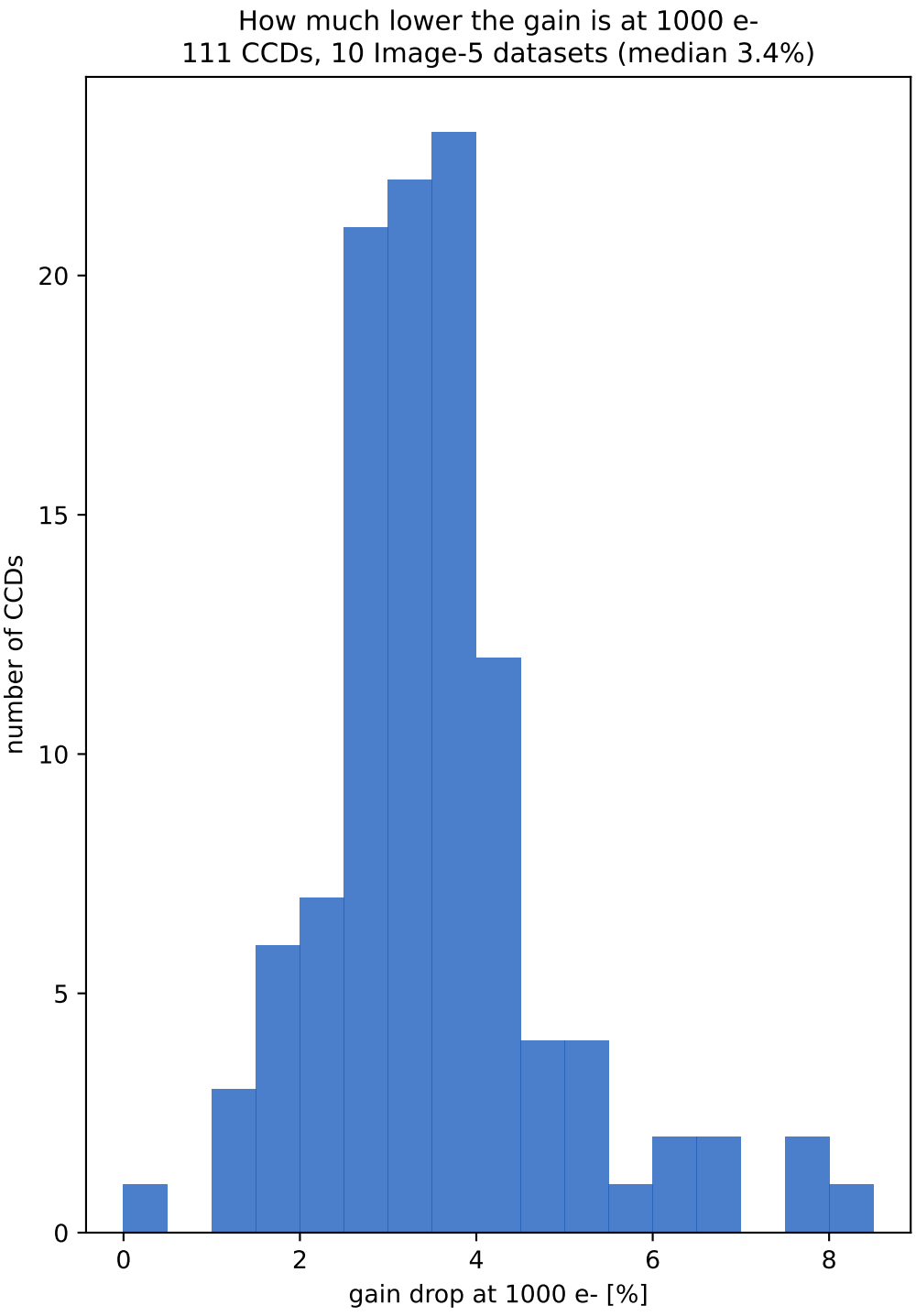
Limits: needs multi-skip images with visible electron peaks (1-skip images are skipped); ~20% of CCDs get no reliable curve and stay as they were; windowed (no-overscan) images improve less; no correction for the Am-241 run (no resolved peaks).

Pages

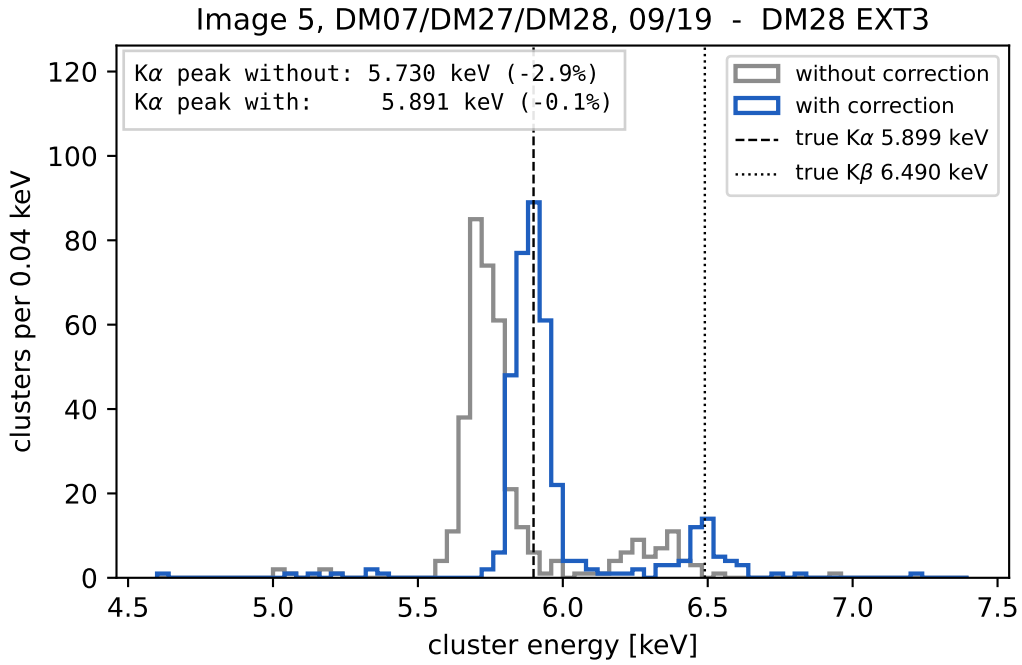
2: measured g(n) curves. 3-5: Fe-55 spectra of single CCDs with/without. 6-8: all CCDs of a module overlaid (alignment). 9: binned image. 10: every dataset, per CCD. 11: limitations.

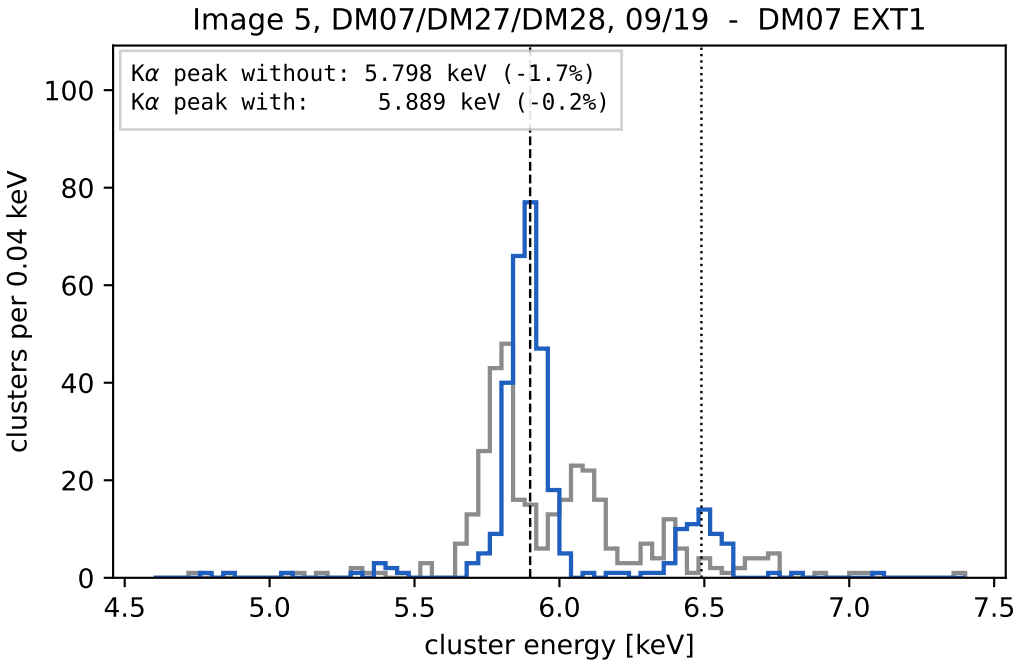
Page 2 - the effect being corrected: the gain falls with the charge in a pixel

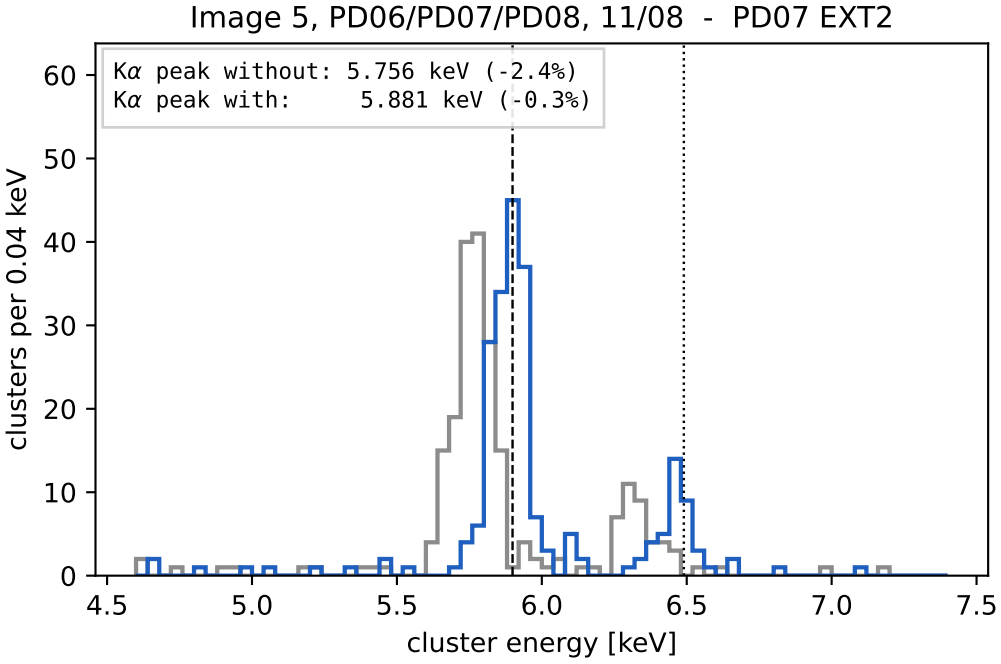


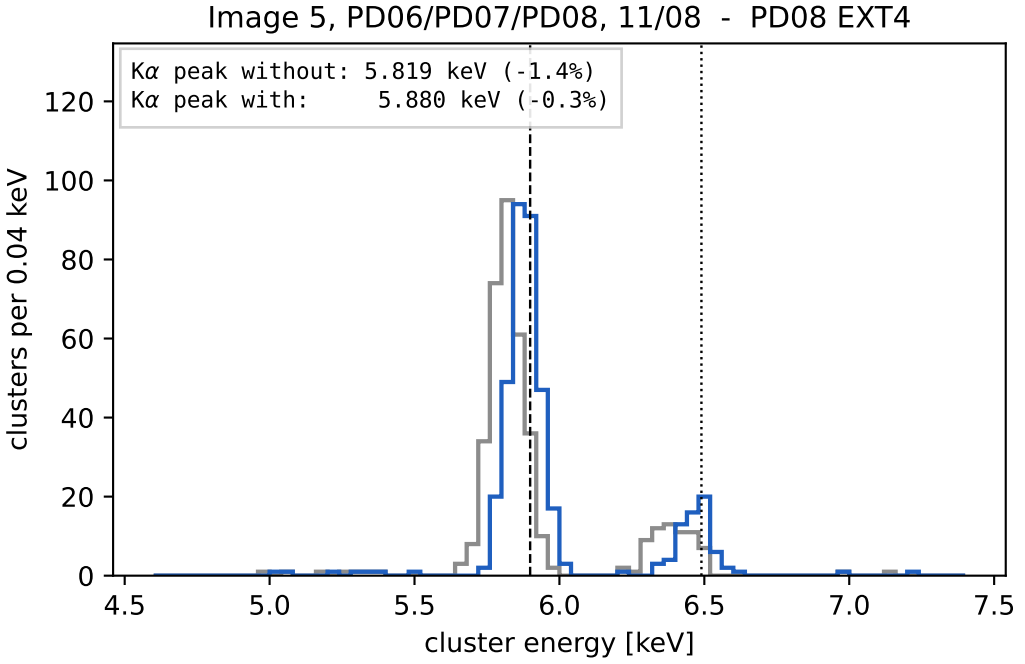


Page 3 - Fe-55 spectrum of single CCDs (front-side clusters)
Typical CCDs: the K-alpha peak moves from ~2% low to the right energy and gets slightly sharper.

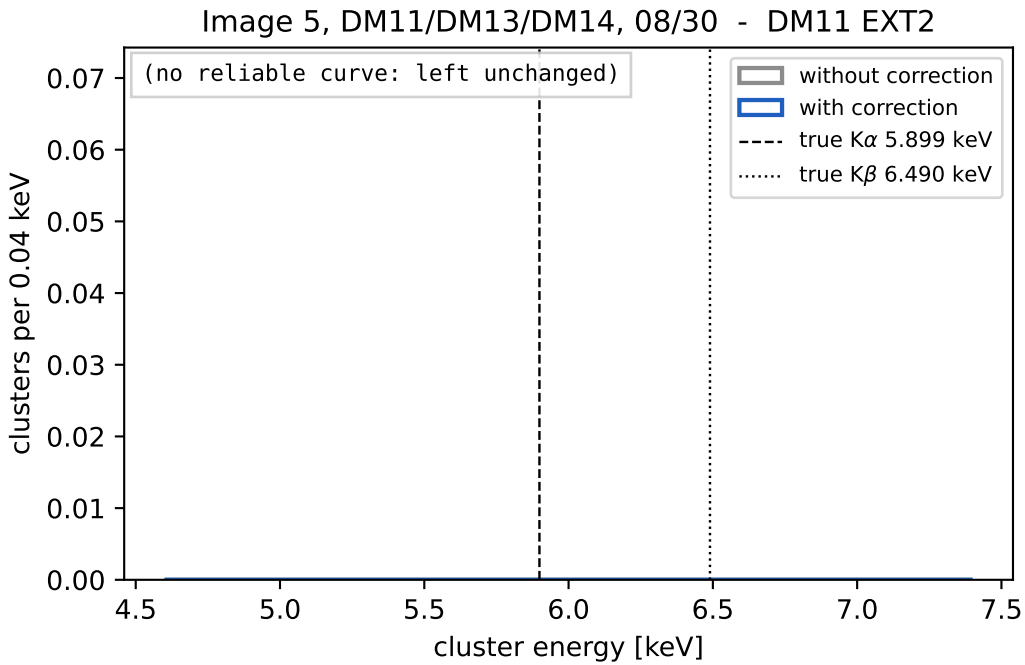


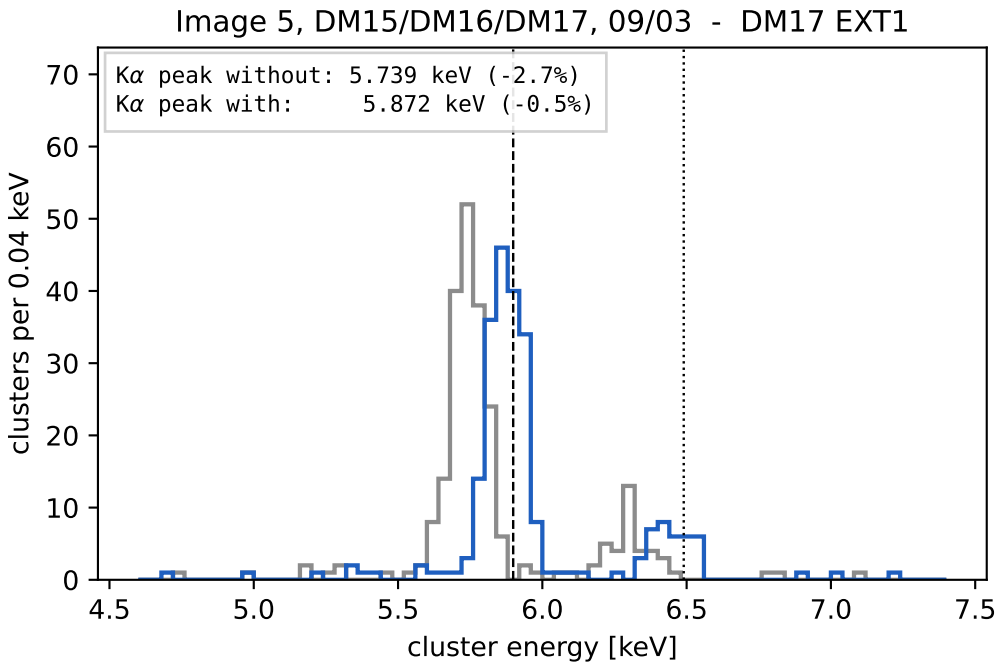


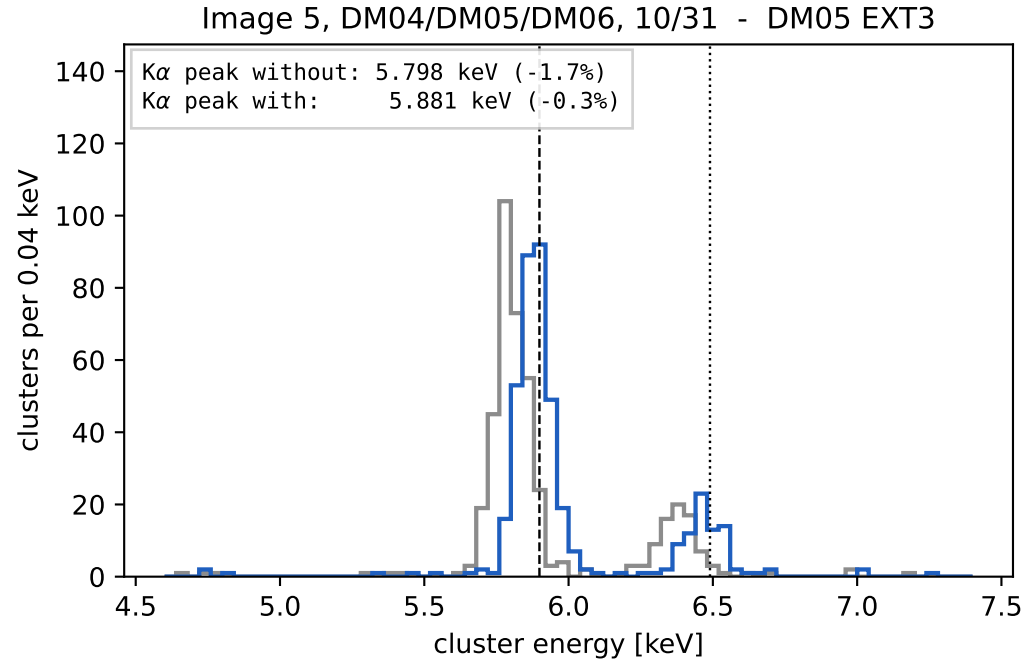


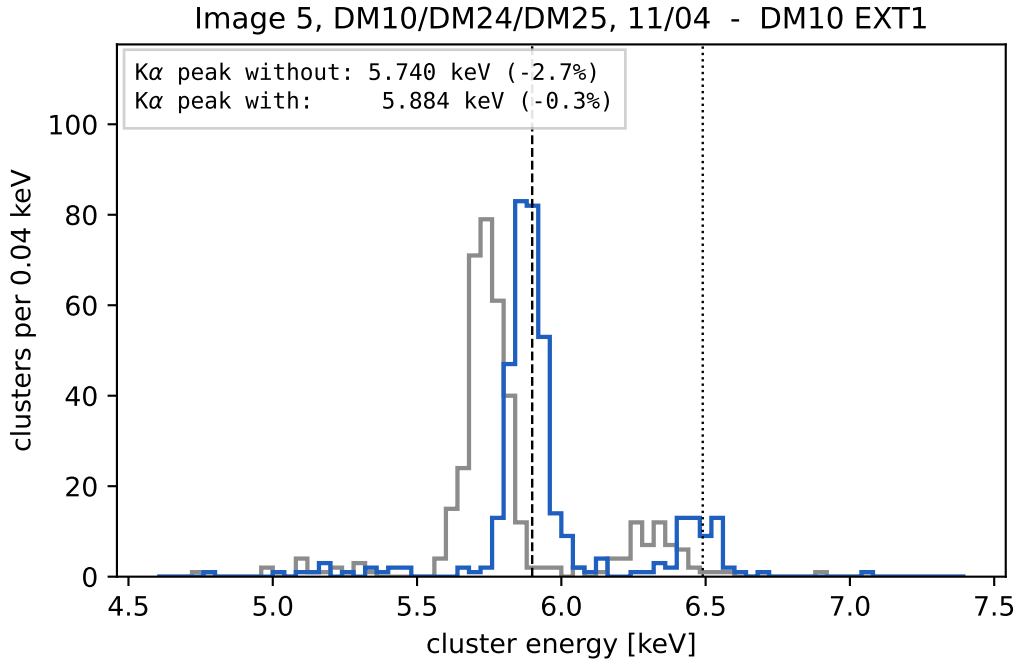


Page 4 - Fe-55 spectrum of single CCDs (front-side clusters)
More modules and dates: same behaviour.

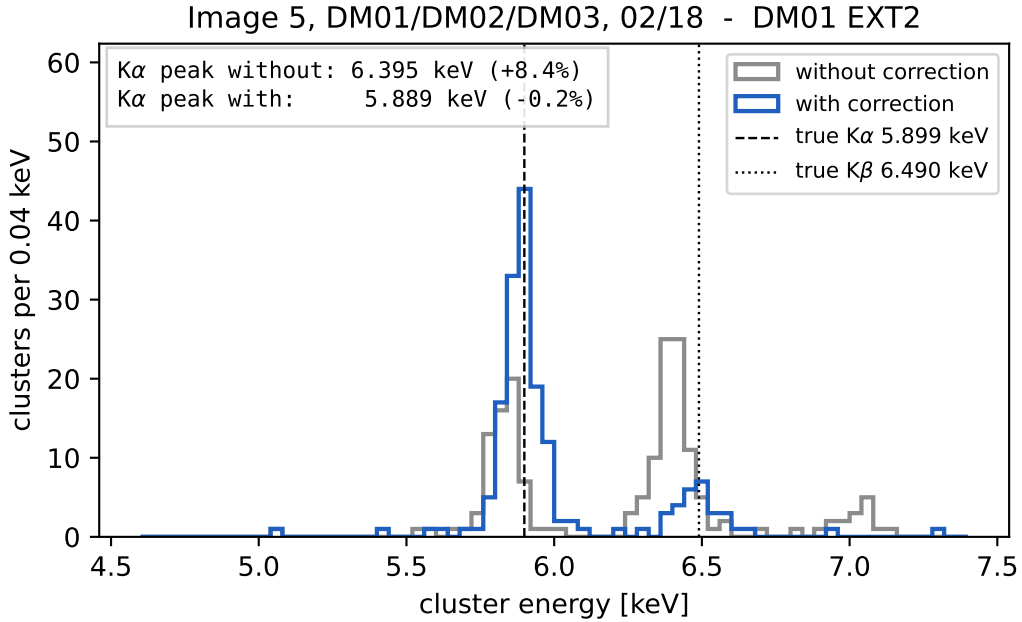


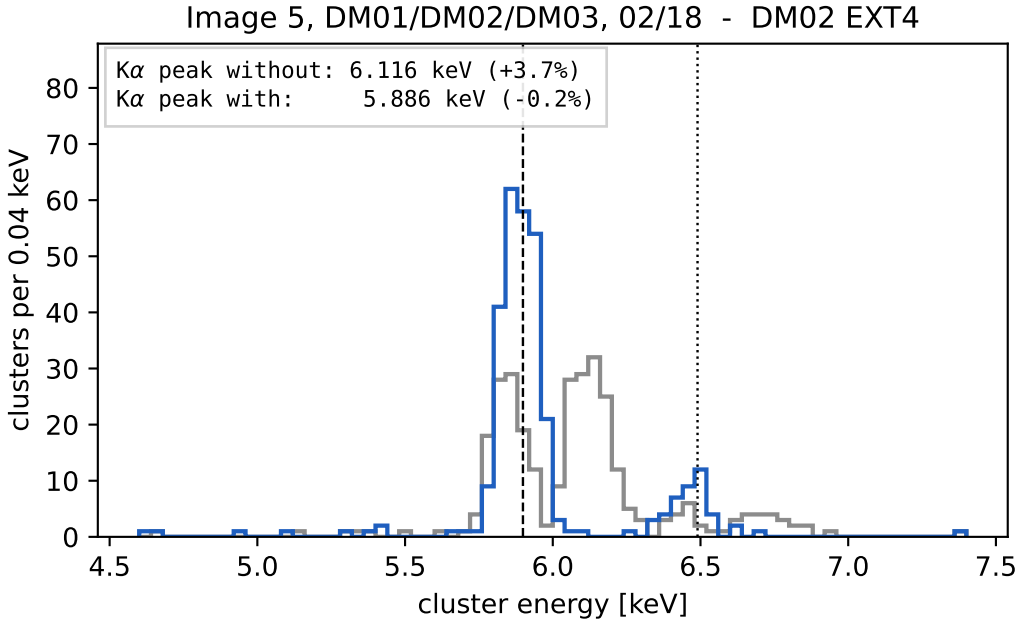


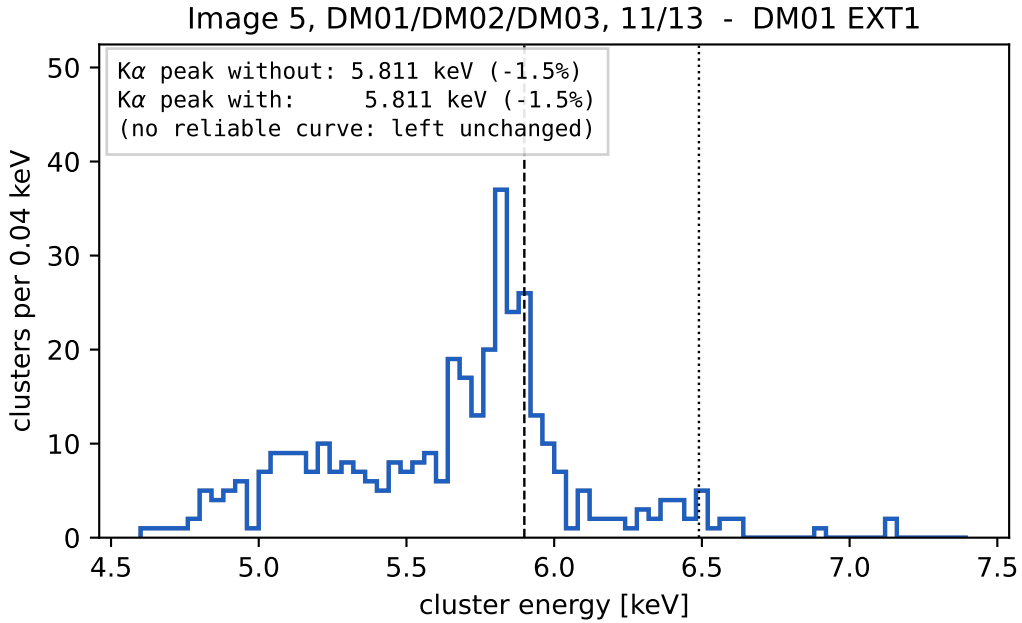


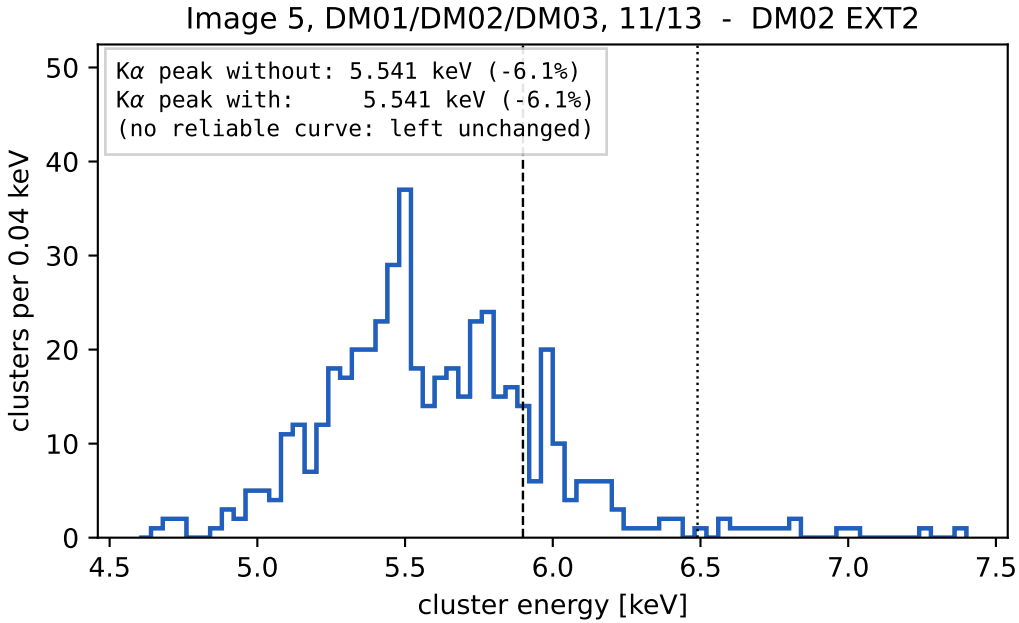


Page 5 - Fe-55 spectrum of single CCDs (front-side clusters)
Top: CCDs that were badly off - the uncorrected peak is shifted or split in two because the per-image fitted gain was wrong or differed between images - come back to one peak at 5.90 keV. Bottom: CCDs without a reliable curve are left unchanged (grey and blue lie on top of each other).

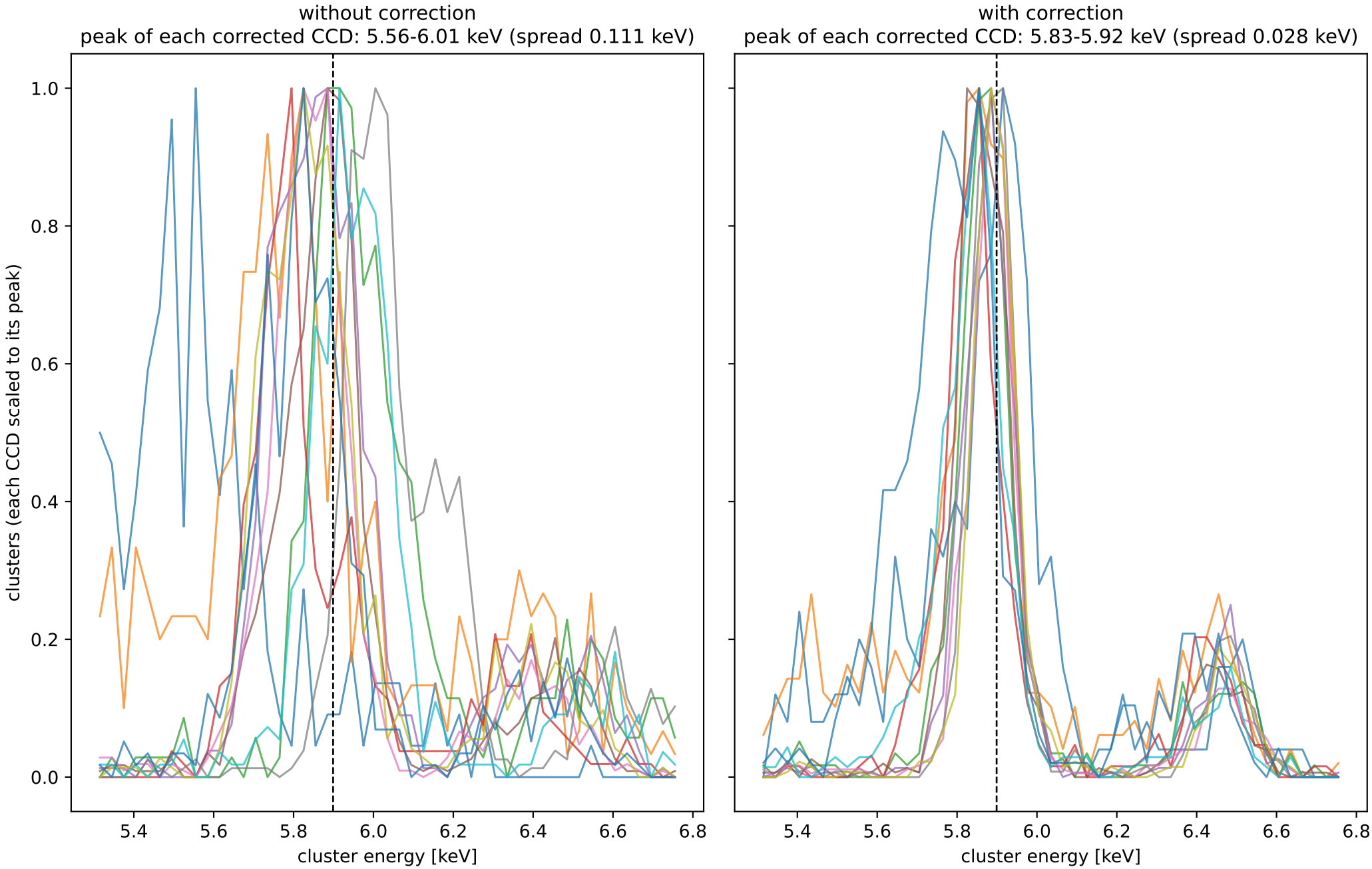




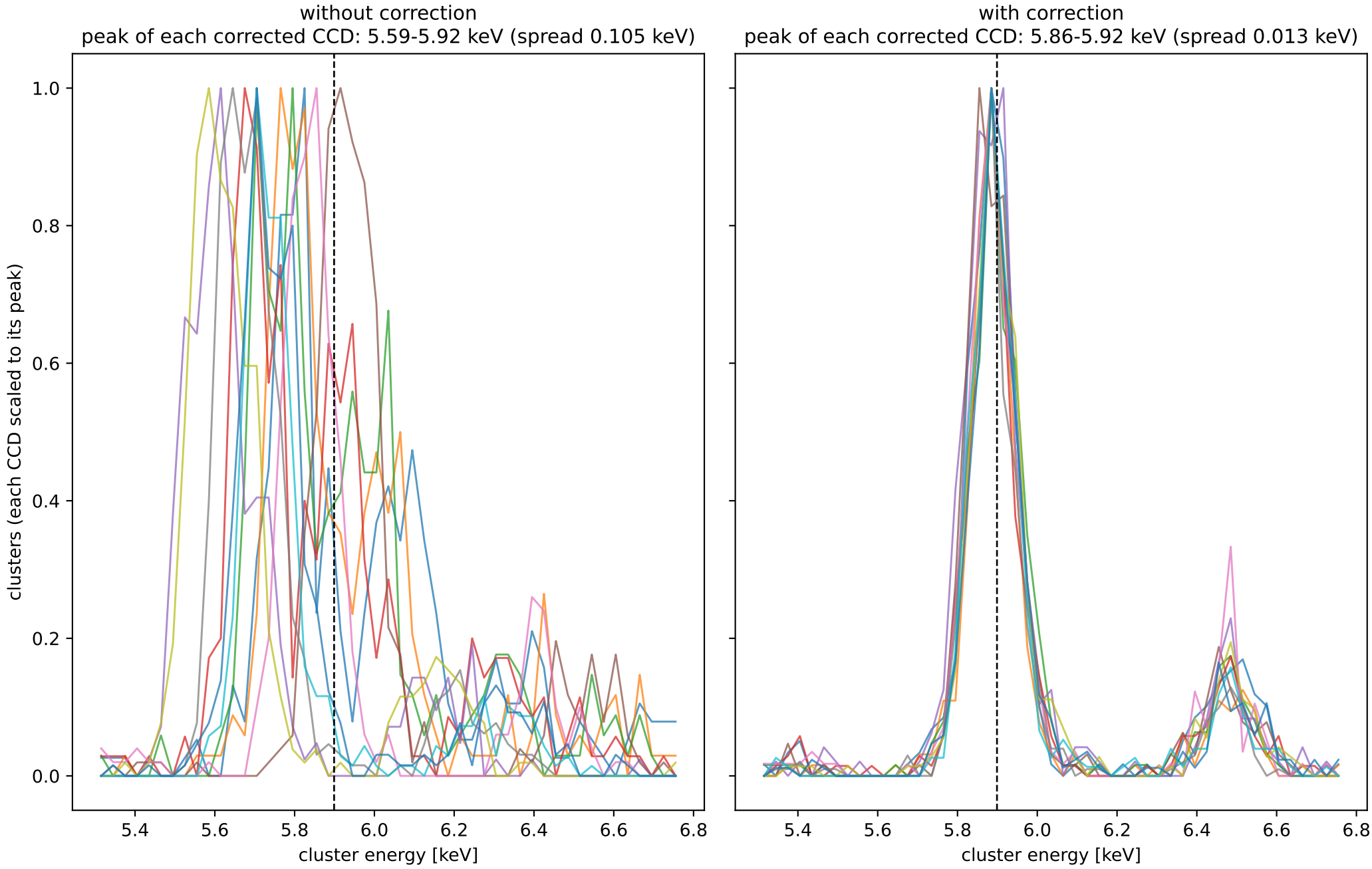




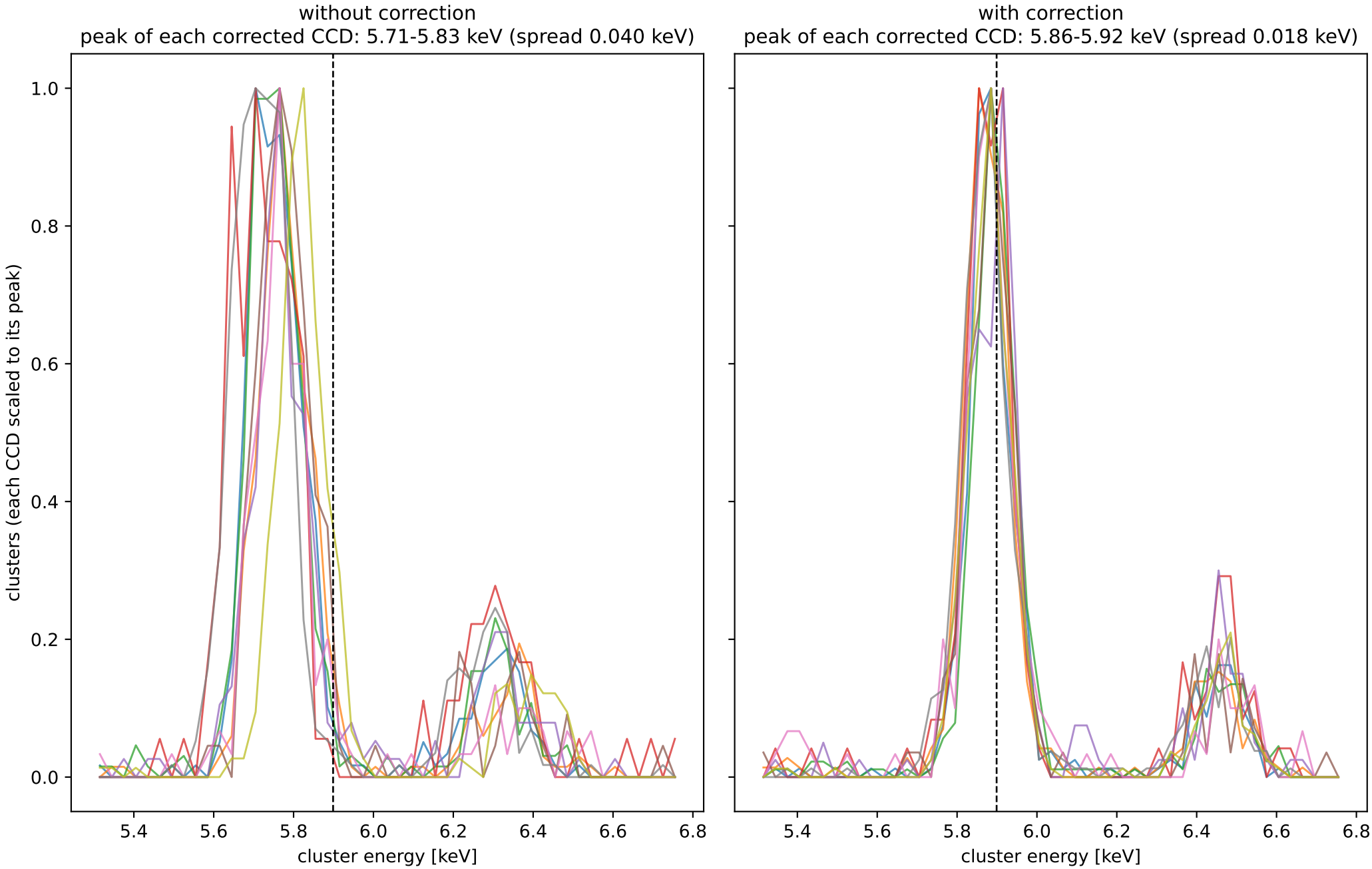
Page 6 - all 12 CCDs of one module set, K-alpha region (front-side clusters)
Image 5, DM24/DM25/DM26, 09/13 (the first test set, 18 images): one line per CCD; black dashed = 5.899 keV;
grey dashed = CCD without a reliable curve (left unchanged, not in the peak range)



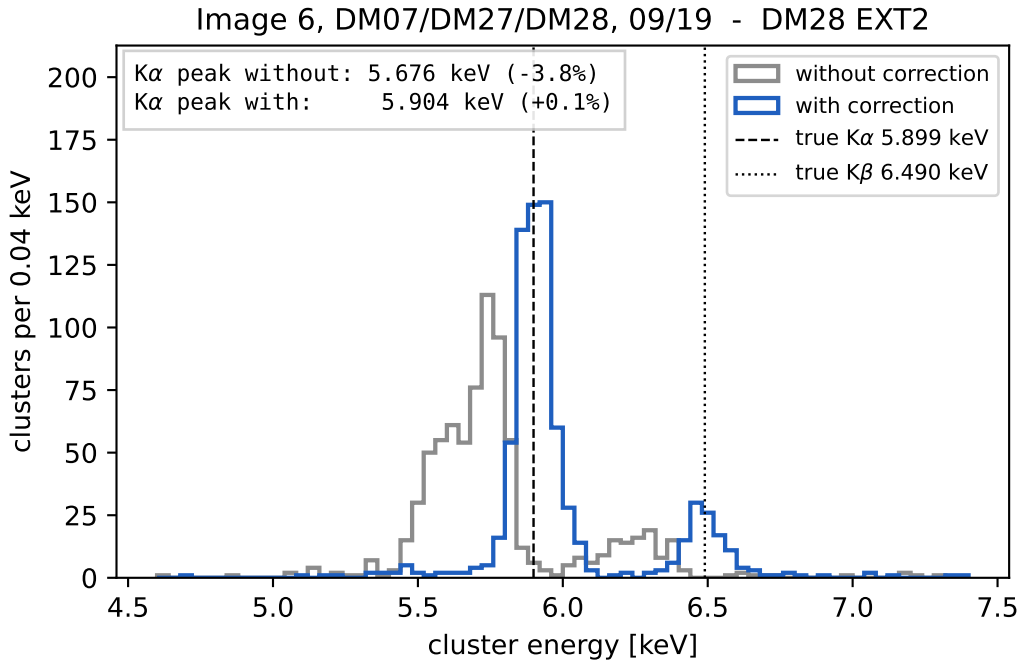
Page 7 - all 12 CCDs of one module set, K-alpha region (front-side clusters)
Image 5, DM07/DM27/DM28, 09/19: one line per CCD; black dashed = 5.899 keV;
grey dashed = CCD without a reliable curve (left unchanged, not in the peak range)

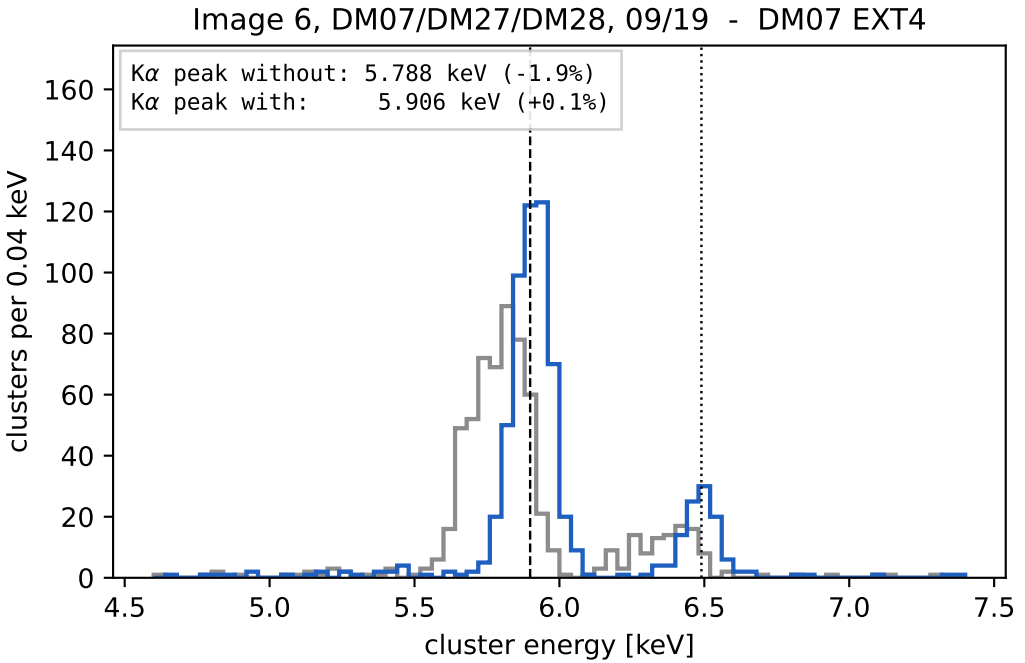


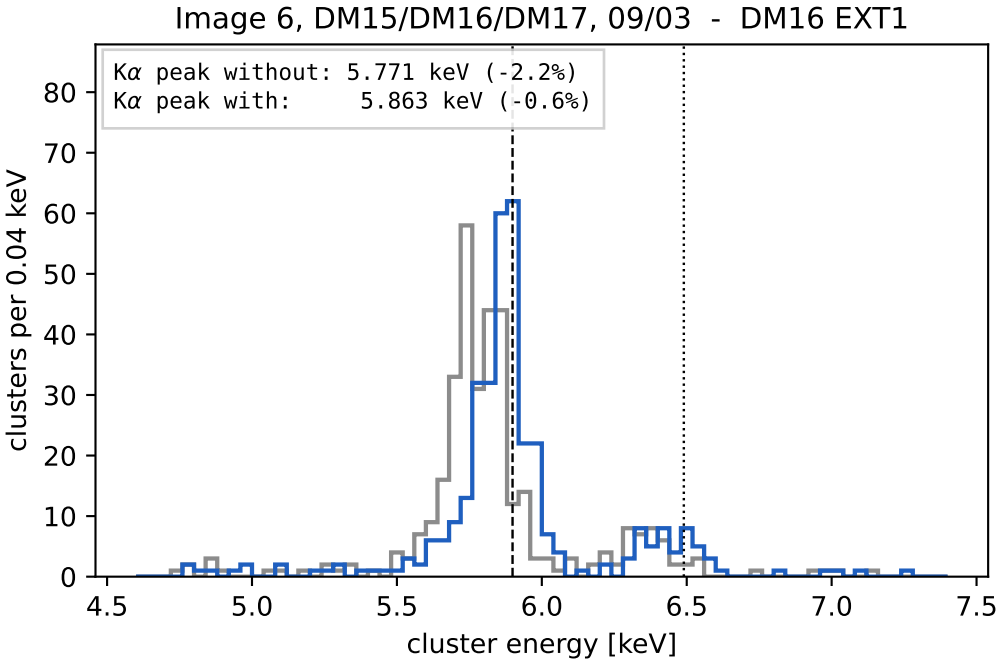
Page 8 - all 12 CCDs of one module set, K-alpha region (front-side clusters)
Image 5, PD06/PD07/PD08, 11/08: one line per CCD; black dashed = 5.899 keV;
grey dashed = CCD without a reliable curve (left unchanged, not in the peak range)

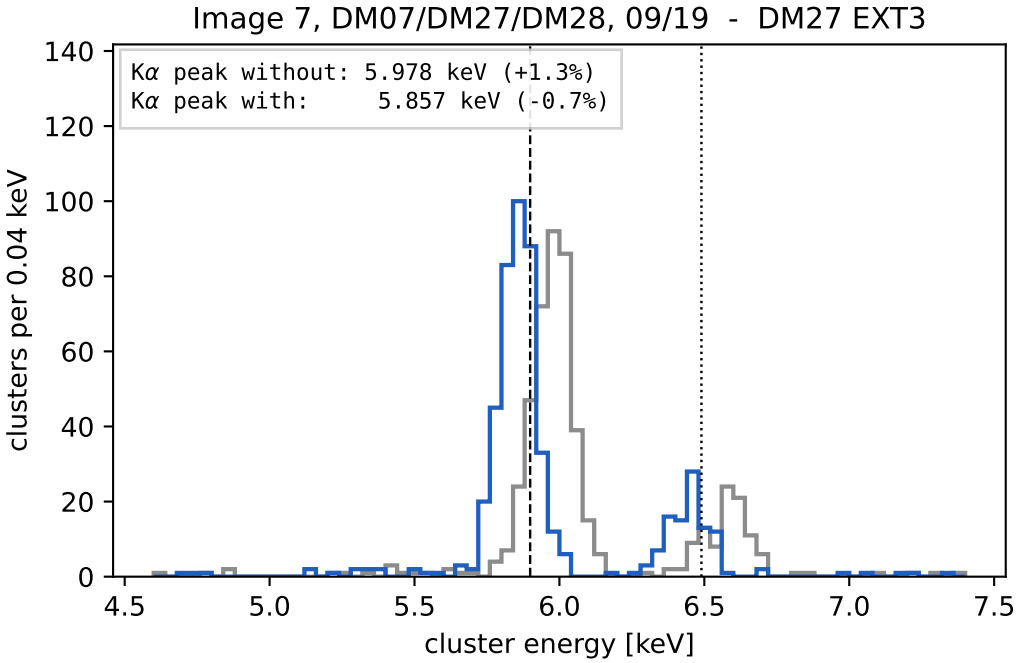


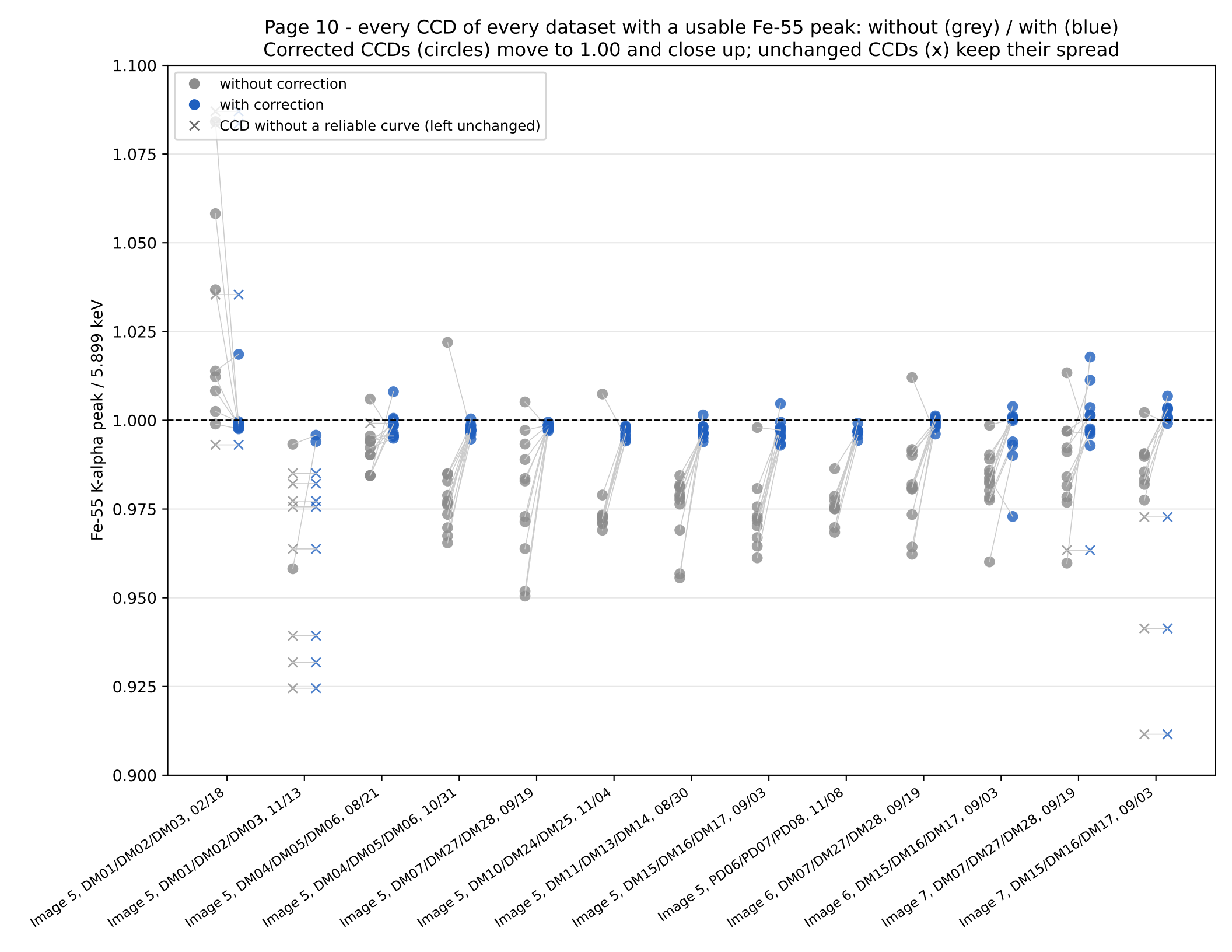
Page 9 - binned images (Image 6: 10 rows summed, Image 7: 10 columns summed; all clusters)
Pixels hold more charge here, so the correction is larger










Page 10 - every CCD of every dataset with a usable Fe-55 peak: without (grey) / with (blue)
Corrected CCDs (circles) move to 1.00 and close up; unchanged CCDs (x) keep their spread
without correction
with correction
CCD without a reliable curve (left unchanged)
Fe-55 K-alpha peak / 5.899 keV
1.100
1.075
1.050
1.025
1.000
0.975
0.950
0.925
0.900
Image 5, DM01/DM02/DM03, 02/18
Image 5, DM01/DM02/DM03, 11/13
Image 5, DM04/DM05/DM06, 08/21
Image 5, DM04/DM05/DM06, 10/31
Image 5, DM07/DM27/DM28, 09/19
Image 5, DM10/DM24/DM25, 11/04
Image 5, DM11/DM13/DM14, 08/30
Image 5, DM15/DM16/DM17, 09/03
Image 5, PD06/PD07/PD08, 11/08
Image 6, DM07/DM27/DM28, 09/19
Image 6, DM15/DM16/DM17, 09/03
Image 7, DM07/DM27/DM28, 09/19
Image 7, DM15/DM16/DM17, 09/03

When the correction cannot help

CCDs without a reliable curve (~20%)

38 of 190 CCD curves in this test were rejected: the electron comb faded after 1-3 charge bins (noisy or charged CCDs), or the measured gain rose with charge. These CCDs are copied unchanged, so a dataset where many are rejected (e.g. DM01-03, 11/13: 10 of 12) barely improves.

Images it does not apply to

1-skip images (no single-electron peaks), CCDs with no resolved peaks (the Am-241 run of 03/28 here), and July 2025 tests with almost no Fe-55 signal (nothing to check against).

Windowed readouts (no overscan, July-August 2025)

Supported since this test, but the improvement is smaller (spread 0.047 -> 0.037 and 0.059 -> 0.050 keV) and a few CCDs get worse; treat these with care.

What remains

Corrected K-alpha sits ~0.2-0.7% below 5.899 keV: charge in the faint edge of each cluster that the clustering threshold drops. A gain correction cannot recover it. Back-side (diffuse) clusters lose more.